Summary

1. Introduction

Customer churn is one of the most critical challenges for telecom companies. Retaining existing customers is often more cost-effective than acquiring new ones, and understanding the reasons behind churn can directly improve profitability and customer satisfaction.

In this project, Exploratory Data Analysis (EDA) is performed on a telecom customer churn dataset to identify the key factors that influence whether a customer decides to stay with the company or leave (churn). Various charts and visualizations have been used to understand patterns across demographics, service usage, contracts, billing, and payment behavior.

2. Objectives of the Study

The main objectives of this EDA are:

- To understand the overall churn rate in the dataset.
- To analyze how customer characteristics (such as senior citizen status, tenure, and charges) affect churn.
- To study the relationship between different services (like Internet, OnlineSecurity, Backup, TechSupport, Streaming etc.) and churn.
- To examine the impact of contract type and payment methods on customer churn.
- To derive actionable insights and recommendations that can help reduce churn and improve customer retention.

3. Dataset Description and Methodology

The dataset represents telecom customers and contains information about:

- **Demographic details**: gender, senior citizen status, partner, dependents.
- **Account details**: tenure (how long the customer has stayed), contract type, billing type, payment method.
- **Service details**: phone service, multiple lines, internet service (DSL/Fiber/None), online security, online backup, device protection, tech support, streaming TV, streaming movies.
- **Target variable**: **Churn** – whether the customer left the company (Yes) or stayed (No).

Methodology:

1. **Data loading and inspection** – checking structure, data types, missing values.
2. **Data cleaning** – handling blanks, converting columns to correct data types.
3. **Univariate analysis** – countplots and distributions for individual variables.

4. **Bivariate analysis** – comparing features with churn using countplots, stacked bar charts, and percentage-based charts.
5. **Insight generation** – interpreting trends and patterns visible in the charts.

4. Exploratory Data Analysis

4.1 Overall Churn Level

The analysis shows that roughly **one-fourth of customers (around 25–30%) have churned**, while the remaining **70–75% have stayed** with the company. This churn rate is significant and indicates a need for focused retention strategies.

4.2 Demographic Insights

- **Senior Citizens:**
  Senior citizens have a noticeably **higher churn rate (around 35–40%)** compared to non-senior customers (around 20–22%).
  This suggests that older customers may be more sensitive to price, service quality, or complexity of plans.
- **Dependents / Family:**
  Customers without dependents or family ties tend to churn slightly more than those with partners or dependents, indicating that family usage may create stickiness and loyalty.

4.3 Service Usage and Add-On Features

Multiple service-related variables were visualized such as **PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, and StreamingMovies**.

Key patterns observed:

- **Internet Service:**
  - Customers with **Fiber Optic internet** show a **much higher churn rate (around 35–40%)** compared to those with DSL (around 15–18%).
  - Customers with **no internet service** have the lowest churn, which is expected as they use fewer services.
- **Add-On Services (OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport):**
  - For almost all these services, **customers who do not have the service churn significantly more (around 25–35%)**.
  - Customers who **do have these services usually show lower churn (around 10–15%)**. This indicates that bundled services improve perceived value and reduce the chance of churn.
- **Entertainment Services (StreamingTV, StreamingMovies):**

    - These services also show different churn rates, but the strongest signals still come from security, backup, and support-related services.

4.4 Contract Type and Payment Behavior

- **Contract Type:**
    - **Month-to-month contracts** have the **highest churn**, often around **45–50%**.
    - **One-year contracts** show a moderate churn rate (around 10–15%).
    - **Two-year contracts** have the **lowest churn (around 2–5%)**.
      This clearly shows that longer-term contracts are associated with higher customer loyalty.
- **Payment Method:**
    - Customers paying via **electronic check** have the **highest churn rate (around 40–45%)**.
    - Those using **credit card, bank transfer (automatic)** have a much lower churn rate (around 15–20%).
    - **Mailed check** users fall in between (around 20–25%).
      This suggests that customers paying electronically without automation may be less satisfied or more price-conscious.

4.5 Tenure and Monthly Charges

- **Tenure:**
    - New customers with **low tenure (less than 1 year)** show a **high churn rate (around 40–45%)**.
    - As tenure increases, churn decreases significantly, with long-term customers having churn rates below 10%.
      This indicates that the first few months are critical for building loyalty.
- **Monthly Charges:**
    - Customers with **high monthly charges** (upper price range) are more likely to churn (around 35–40%).
    - Lower-charge customers have lower churn (around 10–15%).
      This suggests that pricing and bill value are important drivers of customer decisions.

5. Key Insights

- Customers on **month-to-month contracts**, with **electronic check payments**, and **fiber optic internet** show the **highest churn levels**.

- The **absence of add-on services** like OnlineSecurity, OnlineBackup, DeviceProtection, and TechSupport is strongly associated with **higher churn (25–35%)**.
- **Senior citizens, new customers (low tenure), and high monthly charge customers** are high-risk segments.
- Longer contract duration and automatic payments are associated with **significantly lower churn**.

6. Recommendations (Short)

Based on the analysis:

- Encourage **month-to-month customers** to shift to **1-year or 2-year contracts** using discounts, loyalty rewards, or value packs.
- Improve **fiber-optic service experience** and consider special offers or bill relief for dissatisfied users.
- Actively **promote bundled add-on services** (security, backup, support) with free trials or combo packs, especially to high-risk segments.
- Motivate customers to switch from **electronic checks to automatic payment methods** through cashback or fee waivers.
- Provide **special support and simpler plans** for senior citizens and **extra onboarding support** for new customers.

7. Conclusion

This EDA project successfully identifies the major factors linked to customer churn in a telecom company. The analysis highlights that churn is heavily driven by **contract flexibility, lack of value-added services, high billing, payment method, and early-stage dissatisfaction**.

By acting on these insights—through better contract design, targeted offers, service bundling, and customer support—the company can significantly reduce churn, enhance customer satisfaction, and improve long-term profitability.